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Applied Analytics Project - Week 5 Assignment

Feature Engineering, Data Augmentation, and Dimensionality Reduction:

For this week's assignment, our objective was to expand our feature set by creating new variables from our current data fields and to further refine the structure of our dataset for predictive modeling. We also engineered more efficiency metrics using raw box score statistics and incorporated those features into our matchup-level dataset.

From the raw dataset, we separated the winner's and loser's statistics, standardized the columns to match, and combined them into one team-level dataset. We then grouped the data by Season and TeamID to calculate total season statistics for each team.

Using these totals, we created several new efficiency features that combine multiple variables:

- Field Goal Percentage (FGM / FGA)
- Three-Point Percentage ($\text{FGM3} / \text{FGA3}$)
- Free Throw Percentage (FTM / FTA)
- Effective Field Goal Percentage ($(\text{FGM} + 0.5 \times \text{FGM3}) / \text{FGA}$)
- Assist-to-Turnover Ratio (AST / TO)

These features can summarize a team's performance in a more meaningful way than just raw counts. For example, effective field goal percentage uses both total field goals and three-point field goals to demonstrate a team's shooting efficiency better while assist-to-turnover ratio shows a team's ball control and offensive discipline by combining their passing and turnover behaviors. These engineered features provide a more interpretable representation of team performance compared to raw counting stats alone.

Next, we merged these new efficiency metrics into our matchup dataset for both Team 1 and Team 2. To stay consistent with our modeling approach, we converted them into difference-based features (Team 1 minus Team 2) and created new variables such as

“fgPctDiff”, “threePtPctDiff”, “eFgPctDiff”, and “astToDiff”. These directly measure the relative advantage between the two teams.

We did not generate synthetic data this week. However, the feature engineering process helped strengthen the dataset by adding variables with more quality. Instead of increasing the number of rows, we improved the quality of each observation by including efficiency-based predictors.

In terms of dimensionality reduction, these new features also help simplify the dataset. Since these efficiency metrics combine multiple raw statistics into single variables, it helps reduce redundancy. For example, instead of using FGM, FGA, and FGM3 as separate predictors, effective field goal percentage captures information from all those fields. Also, using difference-based features instead of separate team statistics cuts the number of predictors in half and helps reduce multicollinearity.

In summary, this week’s work focused on creating efficiency-based features from multiple raw statistics, merging them into the matchup dataset, and converting them into difference variables. This improved the structure and quality of our predictors while keeping the dataset clean and aligned with our goal of predicting NCAA tournament outcomes.